

제 13강: 선형대수

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▶ 강의 13 – 선형대수(linear algebra)

» 2, 3차 행렬식

• (1) 2차 행렬

① 행렬식(Determinant)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \text{Det}(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = +ad - bc$$

$$a_{11}a_{22}x_1 + a_{12}a_{22}x_2 = b_1a_{22}$$

$$a_{21}a_{12}x_1 + a_{22}a_{12}x_2 = b_2a_{12}$$

$$(a_{11}a_{22} - a_{21}a_{12})x_1 = b_1a_{22} - b_2a_{12}$$

$$\Rightarrow x_1 = \frac{b_1a_{22} - b_2a_{12}}{a_{11}a_{22} - a_{12}a_{21}}$$

② Cramer의 법칙

$$a_{11}x_1 + a_{12}x_2 = b_1$$

$$a_{21}x_1 + a_{22}x_2 = b_2$$

$$\left[\begin{array}{l} A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\ X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \end{array} \right] \rightarrow AX = b$$

$$\text{Det}(A) \neq 0 \quad \Leftrightarrow, \quad a_{11}a_{22} - a_{12}a_{21} \neq 0$$

$$x_1 = \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}$$

$$x_2 = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}$$

$$x_1 = \frac{b_1a_{22} - b_2a_{12}}{a_{11}a_{22} - a_{12}a_{21}}$$

$$x_2 = \frac{a_{11}b_2 - b_1a_{21}}{a_{11}a_{22} - a_{12}a_{21}}$$

▶ 강의 13 – 선형대수(linear algebra)

» 2, 3차 행렬식

• (2) 3차 행렬

① 행렬식(Determinant)

$$\text{Det}(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \Rightarrow 2!$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\text{Det}(A) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{matrix} \Rightarrow 3! \quad \text{Sarrus 전개}$$

$$\text{Det}(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{21} + a_{11}a_{23}a_{32} + a_{12}a_{21}a_{33}$$

② Cramer의 법칙

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{cases} \rightarrow \mathbf{AX} = \mathbf{b}$$

$$\text{Det}(A) \neq 0$$

$$x_1 = \frac{\begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}}{\text{Det}(A)}$$

$$x_2 = \frac{\begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}}{\text{Det}(A)}$$

$$x_3 = \frac{\begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}}{\text{Det}(A)}$$

▶ 강의 13 – 선형대수(linear algebra)

» 2, 3차 행렬식

• (2) 3차 행렬

① 행렬식(Determinant)

$$\text{Det}(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \Rightarrow 2!$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\text{Det}(A) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \Rightarrow 3! \quad \text{Sarrus 전개}$$

$$\text{Det}(A) = \boxed{a_{11}a_{22}a_{33}} + \boxed{a_{12}a_{23}a_{31}} + \boxed{a_{13}a_{21}a_{32}} - \boxed{a_{13}a_{22}a_{21}} - \boxed{a_{11}a_{23}a_{32}} - \boxed{a_{12}a_{21}a_{33}}$$

$$\Rightarrow a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{21})$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$



$$\Rightarrow a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

▶ 강의 13 – 선형대수(linear algebra)

» 2, 3차 행렬식

- (2) 소행렬식, 여인수

① $A_{ij} = A - (i\text{행}, j\text{열})$

$$A_{11} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad A_{23} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

② $M_{ij} = \mathbf{Det}(A_{ij})$: 소행렬식(Minor)

③ $C_{ij} = (-1)^{i+j} M_{ij} = (-1)^{i+j} \mathbf{Det}(A_{ij})$: 여인수(Cofactor)

$$\Rightarrow a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\mathbf{Det}(A_{11}) = M_{11} \quad \mathbf{Det}(A_{12}) = M_{12} \quad \mathbf{Det}(A_{13}) = M_{13}$$

$$= a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13} = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

▶ 강의 13 – 선형대수(linear algebra)

» 2, 3차 행렬식

• (3) n차 행렬식

① 1행에 의한 여인수 전개

$$\begin{aligned} \mathbf{Det}(A) &= a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n} \\ &\Rightarrow a_{11} M_{11} - a_{12}M_{12} + \cdots + (-1)^{i+j}a_{1n}M_{1n} \end{aligned}$$

② i행에 의한 여인수 전개

$$\mathbf{Det}(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

③ j열에 의한 여인수 전개

$$\mathbf{Det}(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

④ A: 삼각행렬

$$\mathbf{Det}(A) = a_{11}a_{22}a_{33} \cdots a_{nn}$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 5 & 2 & 0 & 0 \\ 8 & 6 & 3 & 0 \\ 10 & 9 & 7 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 5 & 6 & 7 \\ 0 & 2 & 8 & 9 \\ 0 & 0 & 3 & 10 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

▶ 강의 13 – 선형대수(linear algebra)

» 2, 3차 행렬식

• 예제 1

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$M_{21} = \text{Det}(A_{21}) = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} = a_{12}a_{33} - a_{13}a_{32} \quad A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$C_{21} = -M_{21}$$

$$M_{22} = \text{Det}(A_{22}) = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} = a_{11}a_{33} - a_{13}a_{31} \quad A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$C_{22} = M_{22}$$

$$M_{23} = \text{Det}(A_{23}) = \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} = a_{11}a_{32} - a_{12}a_{31} \quad A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$C_{23} = -M_{23}$$

▶ 강의 13 – 선형대수(linear algebra)

» 2, 3차 행렬식

• 예제 2

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 2 & 6 & 4 \\ -1 & 0 & 2 \end{bmatrix}$$

③ 1열 전개

$$\begin{aligned} \text{Det}(A) &= a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} = 1 \begin{vmatrix} 6 & 4 \\ 0 & 2 \end{vmatrix} - 2 \begin{vmatrix} 3 & 0 \\ 0 & 2 \end{vmatrix} - 1 \begin{vmatrix} 3 & 0 \\ 6 & 4 \end{vmatrix} \\ &= 1(6 \cdot 2 - 4 \cdot 0) - 2(3 \cdot 2 - 0 \cdot 0) - 1(3 \cdot 4 - 0 \cdot 6) \\ &= 12 - 12 - 12 = -12 \end{aligned}$$

① Sarrus 전개

$$\begin{aligned} A = \begin{vmatrix} 1 & 3 & 0 \\ 2 & 6 & 4 \\ -1 & 0 & 2 \end{vmatrix} \begin{matrix} 1 & 3 \\ 2 & 6 \\ -1 & 0 \end{matrix} \quad \text{Det}(A) = 1 \cdot 6 \cdot 2 + 3 \cdot 4 \cdot (-1) + 0 \cdot 2 \cdot 0 - 0 \cdot 6 \cdot (-1) - 1 \cdot 4 \cdot 0 - 3 \cdot 2 \cdot 2 \\ \Rightarrow 12 - 12 + 0 - 0 - 0 - 12 = -12 \end{aligned}$$

② 1행 전개

$$\begin{aligned} \text{Det}(A) &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = 1 \begin{vmatrix} 6 & 4 \\ 0 & 2 \end{vmatrix} - 3 \begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix} + 0 \begin{vmatrix} 2 & 6 \\ -1 & 0 \end{vmatrix} \\ &= 1(6 \cdot 2 - 4 \cdot 0) - 3(2 \cdot 2 - 4 \cdot (-1)) + 0 \\ &= 12 - 24 = -12 \end{aligned}$$

▶ 강의 13 – 선형대수(linear algebra)

» 2, 3차 행렬식

• 예제 3

$$A = \begin{bmatrix} -3 & 0 & 0 \\ 6 & 4 & 0 \\ -1 & 2 & 5 \end{bmatrix}$$

③ 1열 전개

$$\begin{aligned} \text{Det}(A) &= a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} = 3 \begin{vmatrix} 4 & 0 \\ 2 & 5 \end{vmatrix} - 6 \begin{vmatrix} 0 & 0 \\ 2 & 5 \end{vmatrix} - 1 \begin{vmatrix} 0 & 0 \\ 4 & 0 \end{vmatrix} \\ &= 3(4 \cdot 5 - 0 \cdot 2) \\ &= -60 \end{aligned}$$

① Sarrus 전개

$$\begin{aligned} A = \begin{vmatrix} -3 & 0 & 0 \\ 6 & 4 & 0 \\ -1 & 2 & 5 \end{vmatrix} \begin{matrix} -3 & 0 \\ 6 & 4 \\ -1 & 2 \end{matrix} \quad \text{Det}(A) &= (-3) \cdot 4 \cdot 5 + 0 \cdot 0 \cdot (-1) + 0 \cdot 6 \cdot 2 - 0 \cdot 4 \cdot (-1) - (-3) \cdot 0 \cdot 2 - 0 \cdot 6 \cdot 5 \\ &\Rightarrow -60 + 0 + 0 - 0 - 0 - 0 = -60 \Rightarrow (-3) \cdot 4 \cdot 5 = -60 \end{aligned}$$

② 1행 전개

$$\begin{aligned} \text{Det}(A) &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = -3 \begin{vmatrix} 4 & 0 \\ 2 & 5 \end{vmatrix} - 0 \begin{vmatrix} 6 & 0 \\ -1 & 5 \end{vmatrix} + 0 \begin{vmatrix} 6 & 4 \\ -1 & 2 \end{vmatrix} \\ &= -3(4 \cdot 5 - 0 \cdot 2) \\ &= -60 \end{aligned}$$

▶ 강의 13 – 선형대수(linear algebra)

» 2, 3차 행렬식

• (4) 정리 1

$$A \xrightarrow{\text{기본행 연산}} B$$

① 소거

$$Det(A) = Det(B)$$

② 교환

$$Det(A) = -Det(B)$$

③ 스칼라배

: 공통인수(약수)로 묶어낸다.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \longleftrightarrow B = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$
$$Det(A) = ad - bc \quad Det(B) = bc - ad$$

$$Det(A) = -Det(B)$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \longleftrightarrow B = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

$$Det(A) = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$Det(B) = a_{11} \begin{vmatrix} a_{32} & a_{33} \\ a_{22} & a_{23} \end{vmatrix} - a_{12} \begin{vmatrix} a_{31} & a_{33} \\ a_{21} & a_{23} \end{vmatrix} + a_{13} \begin{vmatrix} a_{31} & a_{32} \\ a_{21} & a_{22} \end{vmatrix}$$

$$Det(A) = -Det(B)$$

▶ 강의 13 – 선형대수(linear algebra)

» 2, 3차 행렬식

• (4) 정리 1

$$A \xrightarrow[\text{기본행 연산}]{} B$$

① 소거

$$\text{Det}(A) = \text{Det}(B)$$

② 교환

$$\text{Det}(A) = -\text{Det}(B)$$

③ 스칼라배

: 공통인수(약수)로 묶어낸다.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \alpha a_{i1} & \alpha a_{i2} & \cdots & \alpha a_{in} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \longleftrightarrow B = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

$$\begin{aligned} \text{Det}(A) &= \alpha a_{i1} C_{i1} + \alpha a_{i2} C_{i2} + \cdots + \alpha a_{in} C_{in} \\ &= \alpha (a_{i1} C_{i1} + a_{i2} C_{i2} + \cdots + a_{in} C_{in}) \end{aligned}$$

$$\text{Det}(B) = a_{i1} C_{i1} + a_{i2} C_{i2} + \cdots + a_{in} C_{in}$$

$$\Rightarrow \text{Det}(A) = \alpha \text{Det}(B)$$

▶ 강의 13 – 선형대수(linear algebra)

» 2, 3차 행렬식

• 예제 2

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 2 & 6 & 4 \\ -1 & 0 & 2 \end{bmatrix}$$

③ 1열 전개

$$\begin{aligned} \text{Det}(A) &= a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} = 1 \begin{vmatrix} 6 & 4 \\ 0 & 2 \end{vmatrix} - 2 \begin{vmatrix} 3 & 0 \\ 0 & 2 \end{vmatrix} - 1 \begin{vmatrix} 3 & 0 \\ 6 & 4 \end{vmatrix} \\ &= 1(6 \cdot 2 - 4 \cdot 0) - 2(3 \cdot 2 - 0 \cdot 0) - 1(3 \cdot 4 - 0 \cdot 6) \\ &= 12 - 12 - 12 = -12 \end{aligned}$$

① Sarrus 전개

$$\begin{aligned} A = \begin{vmatrix} 1 & 3 & 0 \\ 2 & 6 & 4 \\ -1 & 0 & 2 \end{vmatrix} \quad \begin{matrix} 1 & 3 \\ 2 & 6 \\ -1 & 0 \end{matrix} \quad \text{Det}(A) = 1 \cdot 6 \cdot 2 + 3 \cdot 4 \cdot (-1) + 0 \cdot 2 \cdot 0 - 0 \cdot 6 \cdot (-1) - 1 \cdot 4 \cdot 0 - 3 \cdot 2 \cdot 2 \\ \Rightarrow 12 - 12 + 0 - 0 - 0 - 12 = -12 \end{aligned}$$

② 1행 전개

$$\begin{aligned} \text{Det}(A) &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = 1 \begin{vmatrix} 6 & 4 \\ 0 & 2 \end{vmatrix} - 3 \begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix} + 0 \begin{vmatrix} 2 & 6 \\ -1 & 0 \end{vmatrix} \\ &= 1(6 \cdot 2 - 4 \cdot 0) - 3(2 \cdot 2 - 4 \cdot (-1)) + 0 \\ &= 12 - 24 = -12 \end{aligned}$$

▶ 강의 13 – 선형대수(linear algebra)

» 2, 3차 행렬식

• 예제 3

$$A = \begin{bmatrix} -3 & 0 & 0 \\ 6 & 4 & 0 \\ -1 & 2 & 5 \end{bmatrix}$$

③ 1열 전개

$$\begin{aligned} \text{Det}(A) &= a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} = 3 \begin{vmatrix} 4 & 0 \\ 2 & 5 \end{vmatrix} - 6 \begin{vmatrix} 0 & 0 \\ 2 & 5 \end{vmatrix} - 1 \begin{vmatrix} 0 & 0 \\ 4 & 0 \end{vmatrix} \\ &= 3(4 \cdot 5 - 0 \cdot 2) \\ &= -60 \end{aligned}$$

① Sarrus 전개

$$\begin{aligned} A = \begin{vmatrix} -3 & 0 & 0 \\ 6 & 4 & 0 \\ -1 & 2 & 5 \end{vmatrix} \begin{matrix} -3 & 0 \\ 6 & 4 \\ -1 & 2 \end{matrix} \quad \text{Det}(A) &= (-3) \cdot 4 \cdot 5 + 0 \cdot 0 \cdot (-1) + 0 \cdot 6 \cdot 2 - 0 \cdot 4 \cdot (-1) - (-3) \cdot 0 \cdot 2 - 0 \cdot 6 \cdot 5 \\ &\Rightarrow -60 + 0 + 0 - 0 - 0 - 0 = -60 \Rightarrow (-3) \cdot 4 \cdot 5 = -60 \end{aligned}$$

② 1행 전개

$$\begin{aligned} \text{Det}(A) &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = -3 \begin{vmatrix} 4 & 0 \\ 2 & 5 \end{vmatrix} - 0 \begin{vmatrix} 6 & 0 \\ -1 & 5 \end{vmatrix} + 0 \begin{vmatrix} 6 & 4 \\ -1 & 2 \end{vmatrix} \\ &= -3(4 \cdot 5 - 0 \cdot 2) \\ &= -60 \end{aligned}$$

▶ 강의 13 – 선형대수(linear algebra)

» 2, 3차 행렬식

• 예제 4

$$A = \begin{bmatrix} 2 & 0 & -4 & 6 \\ 4 & 5 & 1 & 0 \\ 0 & 2 & 6 & -1 \\ -3 & 8 & 9 & 1 \end{bmatrix} \longrightarrow \begin{vmatrix} 2 & 0 & -4 & 6 \\ 4 & 5 & 1 & 0 \\ 0 & 2 & 6 & -1 \\ -3 & 8 & 9 & 1 \end{vmatrix} \longrightarrow 2 \begin{vmatrix} 1 & 0 & -2 & 3 \\ 4 & 5 & 1 & 0 \\ 0 & 2 & 6 & -1 \\ -3 & 8 & 9 & 1 \end{vmatrix}$$

$$= 2 \begin{vmatrix} 1 & 0 & -2 & 3 \\ 0 & 5 & 9 & -12 \\ 0 & 2 & 6 & -1 \\ 0 & 8 & 3 & 10 \end{vmatrix} = 2(1 \cdot C_{11} + 0 \cdot C_{21} + 0 \cdot C_{31} + 0 \cdot C_{41}) = 2(1 \cdot C_{11})$$

$$= 2 \begin{vmatrix} 5 & 9 & -12 \\ 2 & 6 & -1 \\ 8 & 3 & 10 \end{vmatrix} = 2(2 \cdot C_{21} + 6 \cdot C_{22} + (-1) \cdot C_{23}) = 4 \cdot C_{21} + 12 \cdot C_{22} - 2 \cdot C_{23}$$

$$= -4 \begin{vmatrix} 9 & -12 \\ 3 & 10 \end{vmatrix} + 12 \begin{vmatrix} 5 & -12 \\ 8 & 10 \end{vmatrix} + 2 \begin{vmatrix} 5 & 9 \\ 8 & 3 \end{vmatrix} = -4(90 + 36) + 12(50 + 96) + 2(15 - 72)$$

$$= 1134$$

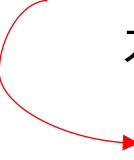
▶ 강의 13 – 선형대수(linear algebra)

» 역행렬, Gauss-Jordan 소거법

- (1) 역행렬

$A = n \times n$ 행렬

- ① 정칙행렬(nonsingular matrix),
가역행렬(invertible matrix)


$$\boxed{A}X = I = XA$$

$X = A^{-1}$: A의 역행렬

- ② 특이행렬(singular matrix)
= 역행렬이 없는 행렬

- (2) <정리 1> $A = n \times n$ 행렬

$$A^{-1} \text{ 존재} \longleftrightarrow \text{Rank}(A) = n$$

$$\text{Rank}(A) = n \longleftrightarrow \text{Det}(A) \neq 0$$

$$A^{-1} \text{ 존재} \longleftrightarrow \text{Det}(A) \neq 0$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \frac{1}{ad - bc}$$

- (3) Gauss-Jordan 소거법으로 역행렬 구하기

$$\boxed{A}X = I$$


$$A^{-1}AX = A^{-1}I \longrightarrow \boxed{I}X = A^{-1}I$$

$$AX = I \longrightarrow [A \mid I]$$

▶ 강의 13 – 선형대수(linear algebra)

» 역행렬, Gauss-Jordan 소거법

• 예제 1

$$A = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$$

$$[A | I] = \begin{bmatrix} -1 & 1 & 2 & 1 & 0 & 0 \\ 3 & -1 & 1 & 0 & 1 & 0 \\ -1 & 3 & 4 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\times 3} \begin{bmatrix} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 7 & 3 & 1 & 0 \\ -1 & 3 & 4 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\times -1} \begin{bmatrix} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 7 & 3 & 1 & 0 \\ 0 & 2 & 2 & -1 & 0 & 1 \end{bmatrix}$$

$$\begin{array}{l} \text{Rank}(A) = 3 \end{array} \xrightarrow{\times -1} \begin{bmatrix} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 7 & 3 & 1 & 0 \\ 0 & 0 & -5 & -4 & -1 & 1 \end{bmatrix} \xrightarrow{\times -\frac{1}{5}} \begin{bmatrix} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 7 & 3 & 1 & 0 \\ 0 & 0 & 1 & \frac{4}{5} & \frac{1}{5} & -\frac{1}{5} \end{bmatrix} \xrightarrow{\begin{matrix} \times -2 \\ \times -7 \end{matrix}} \begin{bmatrix} -1 & 1 & 0 & -\frac{3}{5} & -\frac{2}{5} & \frac{2}{5} \\ 0 & 2 & 0 & -\frac{13}{5} & -\frac{2}{5} & \frac{7}{5} \\ 0 & 0 & 1 & \frac{4}{5} & \frac{1}{5} & -\frac{1}{5} \end{bmatrix}$$

$$\xrightarrow{\times \frac{1}{2}} \begin{bmatrix} -1 & 1 & 0 & -\frac{3}{5} & -\frac{2}{5} & \frac{2}{5} \\ 0 & 1 & 0 & -\frac{13}{10} & -\frac{2}{10} & \frac{7}{10} \\ 0 & 0 & 1 & \frac{4}{5} & \frac{1}{5} & -\frac{1}{5} \end{bmatrix} \xrightarrow{\times -1} \begin{bmatrix} -1 & 0 & 0 & \frac{7}{10} & -\frac{2}{10} & -\frac{3}{10} \\ 0 & 1 & 0 & -\frac{13}{10} & -\frac{2}{10} & \frac{7}{10} \\ 0 & 0 & 1 & \frac{4}{5} & \frac{1}{5} & -\frac{1}{5} \end{bmatrix} \xrightarrow{\times -1} \begin{bmatrix} 1 & 0 & 0 & -\frac{7}{10} & \frac{2}{10} & \frac{3}{10} \\ 0 & 1 & 0 & -\frac{13}{10} & -\frac{2}{10} & \frac{7}{10} \\ 0 & 0 & 1 & \frac{4}{5} & \frac{1}{5} & -\frac{1}{5} \end{bmatrix} = A^{-1}$$

▶ 강의 13 – 선형대수(linear algebra)

» 역행렬, Gauss-Jordan 소거법

- (4) 여인수행렬, 수반행렬

① 여인수행렬

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \quad \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}$$

② 수반행렬(Adjoint)

$$\text{Adj}(A) = (\text{여인수행렬})^T \quad \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \vdots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

▶ 강의 13 – 선형대수(linear algebra)

» 역행렬, Gauss-Jordan 소거법

- (5) <정리 2> $A = n \times n$ 행렬

$$A^{-1} = \frac{1}{\text{Det}(A)} \text{Adj}(A) \quad \text{AX} = I$$

$$\Rightarrow \frac{1}{\text{Det}(A)} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

$$\Rightarrow \frac{1}{\text{Det}(A)} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

$$\Rightarrow \frac{1}{\text{Det}(A)} \begin{bmatrix} a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n} & a_{11}C_{21} + a_{12}C_{22} + \cdots + a_{1n}C_{2n} & \cdots & 0 \\ a_{21}C_{11} + a_{22}C_{12} + \cdots + a_{2n}C_{1n} & a_{21}C_{21} + a_{22}C_{22} + \cdots + a_{2n}C_{2n} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{n1}C_{n1} + a_{n2}C_{n2} + \cdots + a_{nn}C_{nn} \end{bmatrix}$$

▶ 강의 13 – 선형대수(linear algebra)

» 역행렬, Gauss-Jordan 소거법

• 예제 2. $A = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$

$$\text{Det}(A) = 3 \cdot 4 - 1 \cdot 2 = 10$$

$$\text{Adj}(A) = \begin{bmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ -2 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\text{Det}(A)} \text{Adj}(A)$$

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 4 & -1 \\ -2 & 3 \end{bmatrix}$$

• 예제 3. $A = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 3 & -1 \\ -1 & 3 \end{bmatrix}$

$$\text{Det}(A) = 4 - 1 + 18 - 2 + 3 - 12 = 10$$

$$A^{-1} = \frac{1}{10} \begin{bmatrix} \begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix} & -\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} \\ -\begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix} & \begin{vmatrix} -1 & 2 \\ -1 & 4 \end{vmatrix} & -\begin{vmatrix} -1 & 2 \\ 3 & 1 \end{vmatrix} \\ \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} & -\begin{vmatrix} -1 & 1 \\ -1 & 3 \end{vmatrix} & \begin{vmatrix} -1 & 1 \\ 3 & -1 \end{vmatrix} \end{bmatrix}$$

$$A^{-1} = \frac{1}{10} \begin{bmatrix} -7 & 2 & 3 \\ -13 & -2 & 7 \\ 8 & 2 & -2 \end{bmatrix} \begin{bmatrix} -\frac{7}{10} & \frac{2}{10} & \frac{3}{10} \\ \frac{13}{10} & -\frac{2}{10} & \frac{7}{10} \\ -\frac{8}{10} & -\frac{2}{10} & \frac{2}{10} \end{bmatrix}$$

▶ 강의 13 – 선형대수(linear algebra)

» 역행렬, Gauss-Jordan 소거법

- (6) 역행렬의 성질

- ① A: 대각행렬

- (a) $\text{Det}(A) = a_{11}a_{22} \cdots a_{nn}$

- (b) $A = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}$

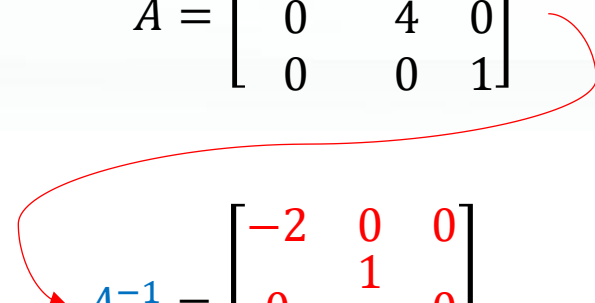
- ② A, B: 정칙행렬

- (a) $AB = \text{정칙행렬}$

- (b) $(AB)^{-1} = B^{-1}A^{-1}$

- 예제 4.

$$A = \begin{bmatrix} -0.5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$


$$A^{-1} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

▶ 강의 13 – 선형대수(linear algebra)

» 역행렬, Gauss-Jordan 소거법

• (7) <정리 3> A, B, C: $n \times n$ 행렬

① $\text{Rank}(A) = n, AB = AC \Rightarrow B = C$

$$\text{Rank}(A) = n \longleftrightarrow A^{-1}: \text{존재}$$

$$A^{-1}AB = A^{-1}AC \Rightarrow IB = IC$$

② a) $\text{Rank}(A) = n, AB = 0 \Rightarrow B = 0$

b) $AB = 0, A \neq 0, B \neq 0 \Rightarrow \text{Rank}(A) < n$

$$A^{-1}AB = A^{-1}0 \Rightarrow IB = 0 \Rightarrow \text{Rank}(B) < n$$

$$\text{Rank}(A) \leq n \begin{cases} \text{Rank}(A) = n \\ \text{Rank}(A) < n \end{cases}$$

③ A: 특이행렬, $AB = \text{특이행렬}$

$$\longleftrightarrow \text{Rank}(A) \neq n$$

$$\longleftrightarrow \text{Det}(A) = 0$$

$$\Rightarrow \text{Det}(AB) = \text{Det}(A) \cdot \text{Det}(B)$$

▶ 강의 13 – 선형대수(linear algebra)

» 역행렬, Gauss-Jordan 소거법

- 예제 6.

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ -1 & 0 & 4 \end{bmatrix}$$

$$A^{-1} = ?$$

$$A^{-1} = \begin{bmatrix} 0.8 & 0 & -0.2 \\ -0.4 & 1 & -0.4 \\ 0.2 & 0 & 0.2 \end{bmatrix}$$

행복한 6월 되세요 ☺

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